

17th March, 2026 (Tue)

Assignment on physics of smartphone

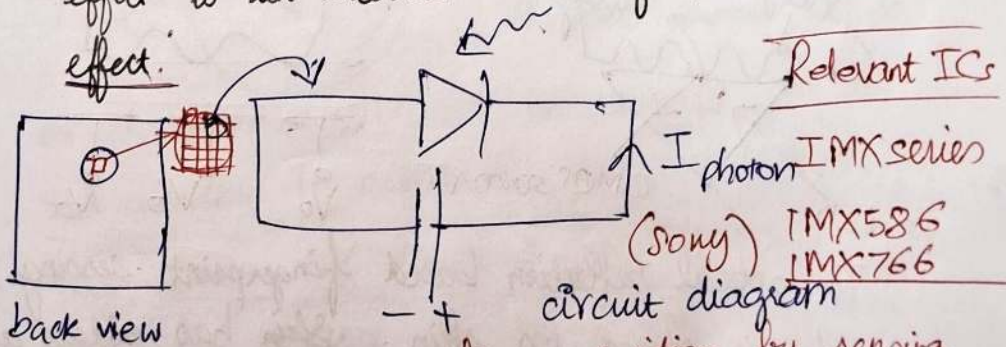
Sensors

① A smartphone camera sensor converts incident light into electric signals. Which physics concept best explains this?

(a) Photoelectric effect (b) Compton scattering (c) Diffraction
(d) Thermionic emission

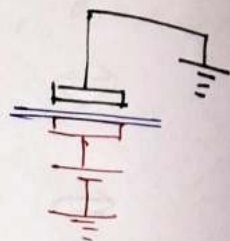
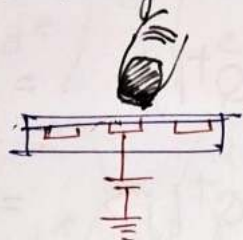
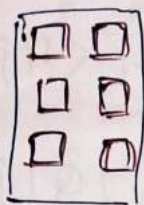
The present technology uses CMOS (Complementary Metal Oxide Semiconductor) based image sensors to convert light into electric signals.

The principle of working of CMOS sensor is photoconductive effect. The p-n junction diode is operated in reverse bias. Upon the shining of light, the photons generate electron-hole pairs and the conductivity increases. Photoconductive effect is an internal version of photoelectric effect.



② Touchscreens detect finger position by sensing changes in electric fields. This technology primarily relies on (a) capacitance (b) electro magnetic induction (c) piezo electric effect (d) Hall effect

The touch screen can be modelled as an insulating glass/polymer substrate with an array of transparent conducting electrodes



front top view

plan view
smartphone

capacitor

Along with the underlying electrode, the finger on top of the screen acts as an electrode; these pair of electrodes form a capacitor with the polymer screen acting as a dielectric medium. By changing the position of finger closer/farther from screen, we change the voltage across the plates of the capacitor.

Relevant ICS

AN85951
(Infineon)

$$\Delta V = \frac{\Delta q}{C_{\text{touch}}}$$

ΔV = voltage across the screen

Δq = charge accumulated at electrode

C_{touch} = screen capacitance

③ Automatic screen rotation in smartphones is achieved using

(a) accelerometer
(Newton's laws)

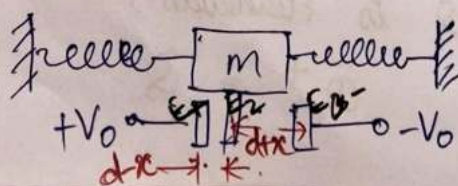
(b) Gyroscope
(Angular momentum)

(c) Both (a) & (b)

(d) Doppler effect

Acceleration of a smartphone is sensed in two ways

(i) Accelerometer



A mass is suspended using springs. The mass has an extension that acts as an electrode E_m . E_m is placed in between two other electrodes E_+ and E_- . E_+ is set to voltage $+V_0$. E_- is set to voltage $-V_0$.

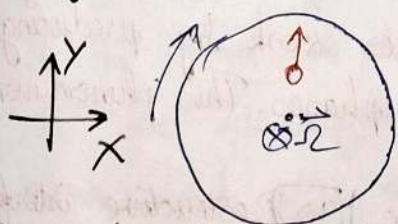
If mass is accelerated at a , the position of E_m changes, thereby the effective capacitance changes. From the change in the capacitance, we can infer the acceleration.

(ii) Gyroscope

The angular rate can be measured by Coriolis acceleration.

What is Coriolis acceleration?

Consider a particle on a rotating platform that is rotating at an angular velocity ω of Ω in clockwise direction. The particle needs



to maintain a no motion along $+Y$ direction.

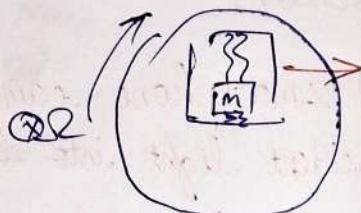
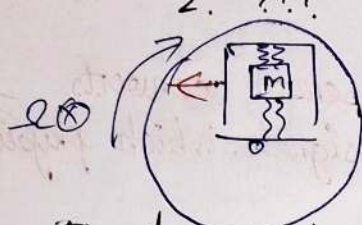
However, since it is

on the platform, it will be dragged to the $+X$ direction. To avoid the drag, it must move along $-X$ direction also. However, as the radial distance increases, the tangential velocity must also keep up, to maintain motion along $+Y$ direction. This increase in velocity is registered as tangential acceleration also called Coriolis acceleration.

$$a_{\text{coriolis}} = -2 \vec{\Omega} \times \vec{v}_{\text{radial}}$$

The factor of 2 is due to

1. maintaining radial motion
2. ???



If the particle moves away from center radially. Here the particle is "tethered". It experiences a force to the right. By reaction, the frame experiences force to the left. By measuring the force, we can infer the angular velocity.

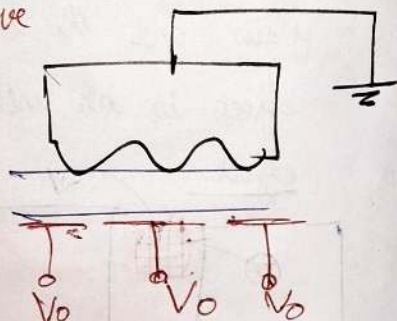
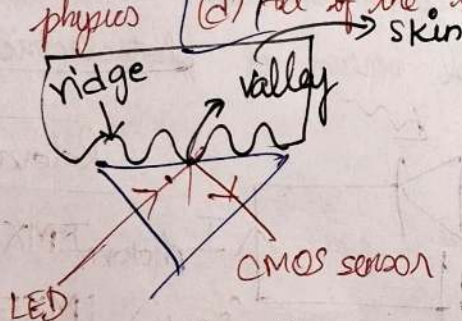
(Analog Devices) Relevant ICs

IC Sensor Gyroscope Accelerometer
is ADXL206

IC Sensor Gyroscope based
is ADXR5645

④ Fingerprint sensors in smartphones operate using

- (a) Optical reflection (b) Electrostatics (c) Semiconductor physics (d) All of the above



In optical reflection based fingerprint sensing, light is shined on skin. Skin has ridges and valleys, leading to differential intensity variations of reflected light. The variations are captured as image using an CMOS image sensor

In the capacitive sensing, the ridge is closer to the conducting electrode below the insulating polymer substrate. It has more capacitance. Whereas, the valley is away leading to low capacitance. The capacitive ~~vari~~ capacitance variations are captured as a grid and stored as fingerprint information. Since we required CMOS image sensor, LED, and touch sensors based on semiconductors, the fingerprint sensor is based on semiconductor physics

Relevant ICs

⑤ Face recognition systems often use infrared depth sensing. The underlying principle involved is (a) Interference of light (b) Time-of-flight measurements of photons (c) Polarization (d) Photoelectric emission.

⑥ Since depth needs to be measured, light needs to be shined on the face. Different features of the face are different depths. The reflected light is measured after a time delay. This is "light echo" similar to a sound echo.

Communication

⑥ Mobile phones communicate with base stations

using

(a) Radio / microwave EM waves (b) IR radiation
(c) visible light (d) X-rays.

The wavelength of EM wave needs to be greater than typical size of source of diffracting objects. Since modern cities have sizes typically of the order of 10m, the wavelengths should be $> 10\text{m}$.

$$\text{So } \nu = \frac{c}{\lambda} = \frac{3 \cdot 10^8}{10\text{m}}$$

$$\sim 0(10^6 - 10^7 \text{ Hz})$$

This falls in the range of 10MHz.

⑦ Mobile signals can bend around buildings due to

(a) Wave nature of EM radiation

(b) particle nature of photons

(c) Relativistic effects

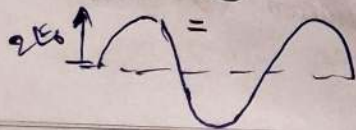
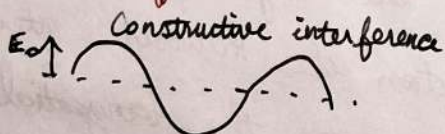
(d) Quantum tunneling

The buildings are sources for diffraction. So EM waves bend due to diffraction

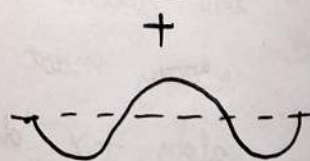
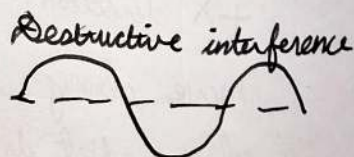
⑧ Noise-cancelling headphones work by producing sound-waves of opposite phases. This phenomenon is known as.

(a) Constructive interference (b) Destructive interference

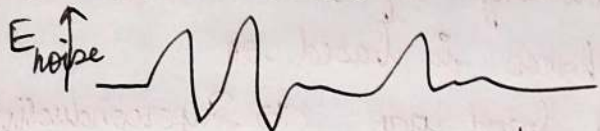
(c) Diffraction (d) Polarization



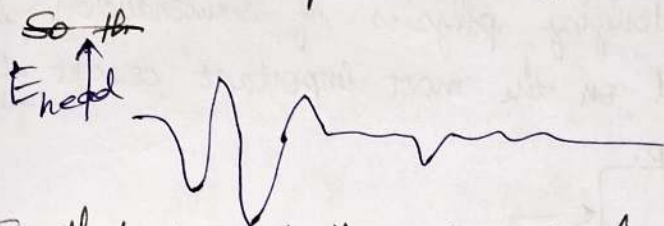
VS



To cancel the noise,



the headphone generates a sound wave that has the same amplitude but opposite in phase

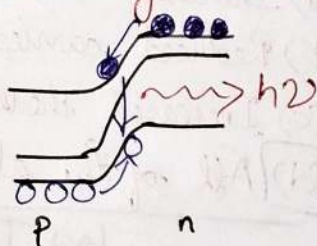
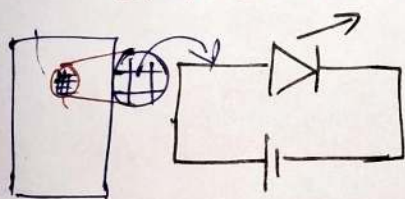


So that sum of the noise signal and headphone generated signal is nearly zero.

Interface

① Light emission in LED and OLED displays occurs due to:

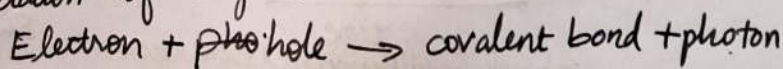
- (a) Incandescence
- (b) Electroluminescence
- (c) Photoelectric effect
- (d) Blackbody radiation



Electroluminescence

is the electrical means to generate luminance.

Under forward bias, the pn junction diode based on direct band gap semiconductor has electrons and from the n-region and holes from the p-region drifting towards each other. When they meet at the depletion layer, recombination of electron-hole pairs leads to the generation of light.



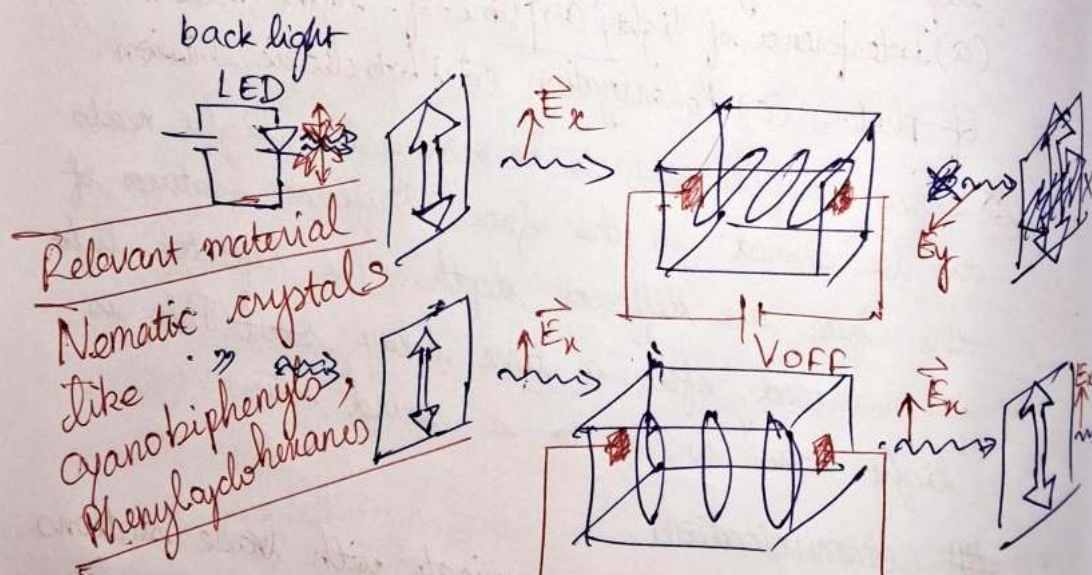
⑥ Change in appearance of phone screens at large viewing angles is mainly due to

- (a) Polarization effects (b) Reflection and refraction
(c) Dispersion (d) Scattering

The state of a pixel in a screen ~~is~~ can take ~~values~~ continuous values from max brightness to minimum (ideally zero) brightness



The brightness is varied by the orientation of liquid crystal molecules with respect to the polarization axis of polarizer.



At large viewing angles vs small viewing angles the polarization changes. The light is not effectively



vs



polarized at large viewing angles, leading to reduction of intensity.

⑪ Blue-light filters in smartphones reduce eye-strain by modifying

(a) Wavelength range (b) Speed of light

(c) Phase of light (d) Polarization only

If the emission spectrum of LCD display is



then blue filter selectively transmits the longer wavelengths. So it adjusts the wavelength range.

Internals

⑫ Proximity sensors in smartphones switch off the display during calls using

(a) Reflection of infra-red radiation

(b) Photo electric effect

(c) Thermal radiation

(d) Doppler shift

Proximity sensor is an IR (emitter + detector) IC that switches off display if IR light is detected.

The emitter is a IR LED.

The detector is a IR photodiode.

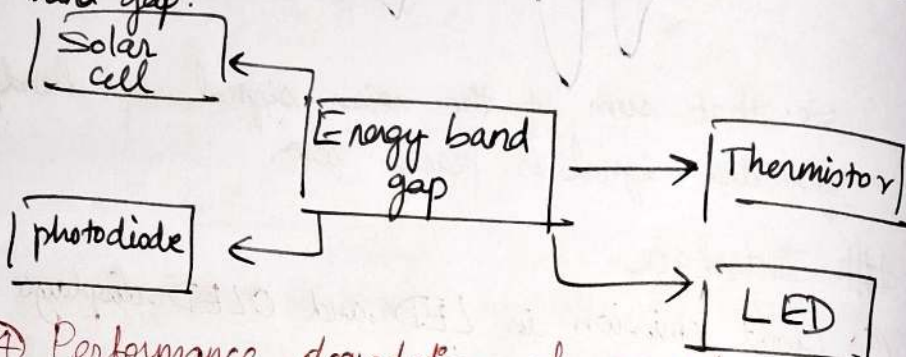
Relevant ICs

AMS OSRAM ICs

⑬ The functioning of semiconductor chips in mobile phones is based on

- (A) Energy band gap (B) Superconductivity
(C) Ferromagnetism (d) Nuclear energy levels

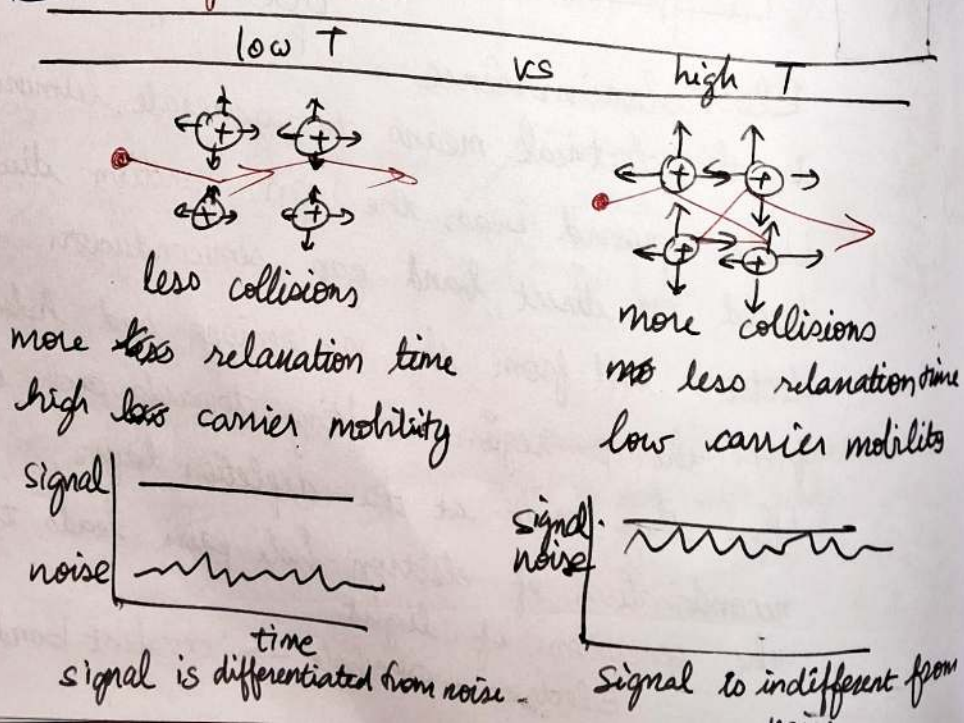
The underlying physics of semiconductor is anchored on the most important concept of energy band gap.



⑭ Performance degradation of smartphones at high temperatures is due to

- (a) Increased lattice vibrations
(B) Reduced carrier mobility
(c) Increased thermal noise

(d) All of the above



Energy

⑫ ⑮ Heat generation during fast charging of mobile phones is governed by

(a) Joule heating ($I^2 R$ loss)

(b) Stefan-Boltzmann law

(c) Wien's displacement law

(d) Faraday's law

Fast charges allow large currents to flow through the phone (typically the power is 25W)

~~If the voltage~~ If the battery capacity is 5000 mAh and it takes 2 hours to charge fully, then current passing on an average is

$$\Delta q = \langle I \rangle \cdot \Delta t$$

$$\Rightarrow 5000 \text{ mAh} = \langle I \rangle \cdot 2 \text{ hr}$$

$$\Rightarrow \langle I \rangle = 2500 \text{ mA} = 2.5 \text{ A}$$

If the power is 25W

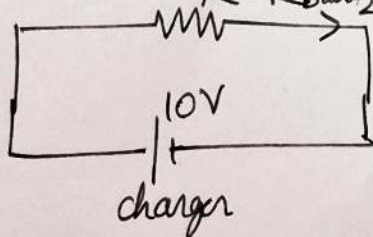
$$\text{then } P = VI$$

$$\Rightarrow V = \frac{P}{I} = \frac{25}{2.5} = 10 \text{ V}$$

So the battery can be modelled as

a resistor with resistance R

$$R = R_{\text{battery}} + R_{\text{charger}}$$



$\Rightarrow$ From Ohm's law

$$R = \frac{V}{I}$$

$$= \frac{10 \text{ V}}{2.5 \text{ A}} = 4 \Omega$$

$\therefore$ Heat dissipated across the battery

is at the rate

$$P = I^2 R_{\text{battery}} = (2.5)^2 \cdot 4 = 25 \text{ W}$$

Joule heating

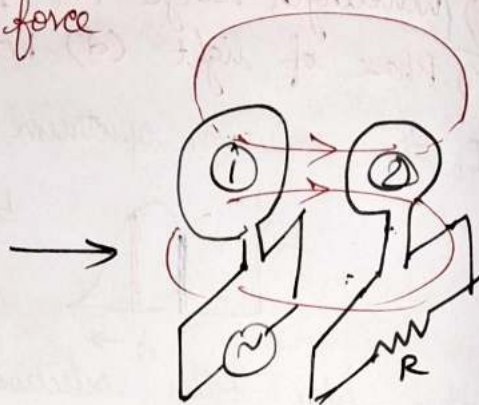
(16) Wireless charging of smartphones works on the principle of

(a) electromagnetic induction

(b) photo electric effect

(c) electrostatic force

(d) Hall effect



Changing electric field
current generates changing magnetic field in circuit 1
and the field lines of loops ① & ② get linked
the changing field generates current in circuit 2
This is the principle of Faraday's law of Electromagnetic induction.

(17) Match the following technologies with the most relevant physics concept

Technology	Physics concept
Camera sensor	Photo-electric effect
Touch screen	Electrostatics (capacitance)
Speaker	Electromagnetic induction
Microphone	Electromagnetic induction
GPS	Time of flight of EM waves
Battery	Electrochemistry